$$ES = \frac{\sigma S}{\varepsilon_0}, \tag{31}$$

$$E = 3E_0 \cos\theta. \tag{32}$$

도체 구와 대전된 고리

2.10 , E
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$$\vec{p} = q\vec{a}, \tag{33}$$

$$q = \rho\frac{4}{3}\pi r^3. \tag{34}$$

(27) (33) (34)

$$\vec{p} = 4\pi r^3 \varepsilon_0 \vec{E}, \tag{35}$$

E

z . .

, Δq_i .

$$\Delta E_z = \frac{\Delta q_i \cos\alpha}{4\pi\varepsilon_0 (z^2+R^2)^2}. \tag{36}$$

$$\cos\alpha = \frac{z}{(z^2+R^2)^{1/2}} \tag{37}$$

$$E(z) = \frac{qz}{4\pi\varepsilon_0 (z^2+R^2)^{3/2}}. \tag{38}$$

$$F = qE(z+a) - qE(z) = qa\frac{dE}{dz} = p\frac{dE}{dz}. \tag{39}$$

(35) (38) (39)

$$F = \frac{q^2 r^3 z(R^2-2z^2)}{4\pi\varepsilon_0 (z^2+R^2)^4}. \tag{40}$$

(40) .

$$z_1 = 0, \tag{41}$$

$$z_{2,3} = \pm\frac{R}{\sqrt{2}}. \tag{42}$$

(41) ,

(42) .

(40)

$$F = -\frac{8q^2 r^3 x}{81\pi\varepsilon_0 R^6}, \tag{43}$$

$$x = z - \frac{R}{\sqrt{2}} \ll R. \tag{44}$$

x ,

$$m\ddot{x} + \frac{8q^2 r^3}{81\pi\varepsilon_0 R^6}x = 0, \tag{45}$$

$$\omega = \sqrt{\frac{8q^2 r^3}{81\pi\varepsilon_0 m R^6}}. \tag{46}$$

2.11 (40) . ,
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